

Data	Type	Designation	
	bin syn model	fiducialCE1	
	engine model	W18-BH2.7-a2.0	
	MbaryLimBH	2.7	Msol
	is2PcritBool	1	
	Progenitor models	{WH15, SW14, WH07}	
	IBD data	Strumia et al. 2003 table 1	
progen+engine	W18-BH2.7-a2.0		
	NS pop without IMF weights	57. %	114
	BH pop without IMF weights	43. %	86
	NS pop IMF weighted	73.0811 %	
	BH pop IMF weighted	26.9189 %	
Progen+engine	Pop single stars		
	Missing BH pop	0	after MbaryLimBH
	total pop	200	
	BH pop	86	43. % not IMF weighted
	NS pop	114	57. % not IMF weighted
2 parameter criterion			
	chosen indep X var	$M4*\mu4$	
	chosen dep Y var	$\mu4$	
	optimal criterion function	$\mu4[M4*\mu4] = 0.32$ $M4*\mu4 + 0.040018$	
	crit outliers with progen+engine	8.5 %	
	max success with progen+engine	91.5 %	
Alternative 1 parameter criterion			
	optimal $\mu4$ threshold (1P alternative)	$\mu4 = 0.089$	
	max success with progen+engine (1P alternative)	88. %	
chosen criterion			
	is2PcritBool	1	
	chosen criterion	$\mu4[M4*\mu4] = 0.32$ $M4*\mu4 + 0.040018$	
	chosen success fraction	0.915	
detectors			
fiducial volumes	detector	kton	
	Super-K Gd	fidVolSK	
	Hyper-K	fidVolHK	
	JUNO	fidVolJ	
number of targets	detector	protons	
	Super-K Gd	$\frac{NtSK}{efficiencySKGd}$	
	Hyper-K	$\frac{NtHK}{efficiencyHK}$	
	JUNO	$\frac{NtJ}{efficiencyJ}$	
efficiency	detector		
	Super-K Gd	efficiencySKGd	
	Hyper-K	efficiencyHK	
	JUNO	efficiencyJ	
	binary synthesis data:	fiducialCE1	
bin syn	Population	Pop binaries	% to total pop
	total pop	478 153	
	all NS pop	478 153	
	NS+BH pop (with crit)	478 153	
	NS pop (with crit)	309 877	64.8071 %
	BH pop (with crit)	168 276	35.1929 %
	outside full bound: total pop outliers	32 389	
	outside NS/BH bound: all NS pop outliers	122 147	25.5456 %
	NS+BH pop (with crit) outliers	32 630	6.82418 %
	NS pop (with crit) outliers	3	0.000627414 %
	BH pop (with crit) outliers	32 627	6.82355 %
	inside full bound: total pop without outliers	445 764	
	inside NS/BH bound: all NS pop without outliers	356 006	74.4544 %
	NS+BH pop (with crit) without outliers	445 523	93.1758 %
	NS pop (with crit) without outliers	309 874	64.8065 %
	BH pop (with crit) without outliers	135 649	28.3694 %
	All NS pop conserved	True	True
	NS+BH pop conserved	True	True
	no binary synthesis data:	fiducialCE1	
no bin syn	Population	Pop single stars	% to total pop
	total pop	416 202	
	all NS pop	416 202	
	NS+BH pop (with crit)	416 202	
	NS pop (with crit)	253 369	60.8764 %
	BH pop (with crit)	162 833	39.1236 %
	outside full bound: total pop outliers	39 991	
	outside NS/BH bound: all NS pop outliers	117 347	28.1947 %
	NS+BH pop (with crit) outliers	40 105	9.63595 %
	NS pop (with crit) outliers	0	0.
	BH pop (with crit) outliers	40 105	9.63595 %
	inside full bound: total pop without outliers	376 211	
	inside NS/BH bound: all NS pop without outliers	298 855	71.8053 %
	NS+BH pop (with crit) without outliers	376 097	90.3641 %
	NS pop (with crit) without outliers	253 369	60.8764 %
	BH pop (with crit) without outliers	122 728	29.4876 %
	All NS pop conserved	True	True
	NS+BH pop conserved	True	True